RICHARD LIU

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EDUCATION

New York University Sept 2024 - Apr 2027 (Expected)
Division of Biostatistics, Department of Population Health, NYU Langone Health
Thesis: Non-parametric Definition, Identification, and Estimation of Causal Effects with Causally Ordered Data
Ph.D. Advisor: Dr. Iván Díaz

University of Michigan, Ann Arbor Sept 2021 - Apr 2023
Department of Biostatistics, School of Public Health
GPA: 4.0/4.0
Courses: Advanced Inference I (Ph.D), Advanced Inference II (Ph.D), Advanced Computational Statistics (Ph.D), Statistical Learning (Ph.D), Stochastic Process, Generalized Linear Model, Statistical Computing, Longitudinal Data Analysis, Survival Analysis, Natural Language Processing (EECS), Deep Learning in Computer Vision (EECS)

Xiamen University Spetember 2016 - June 2020
Department of Informational and Computational Mathematics (Honor Program), School of Mathematics
Major GPA: 3.9/4.0; 91.7/100.0 Rank: 1/28
Courses: Functional Analysis, Numerical Solutions of Partial Differential Equations, Numerical Linear Algebra, Probability, Mathematical Statistics, Regression Analysis, Operations Research, Numerical Optimization
Thesis: An Accelerated Algorithm for Solving Dictionary Learning and Sparse Coding Problem

PUBLICATIONS

1. **Liu R**, Williams NT, Rudolph KE, Diaz I. `crumble`: A comprehensive framework for modern causal mediation analysis with intermediate confounding. arXiv preprint arXiv:2604.09902. 2026 Apr 10. *Under revision at Epidemiology*
2. Vo TT, Williams N, **Liu R**, Rudolph KE, Díaz I. Recanting twins: Addressing intermediate confounding in mediation analysis. *Statistics in Medicine*. 2026 Feb;45(3-5):e70432.
3. Williams NT, **Liu R**, Hoffman KL, Forrest S, Rudolph KE, Díaz I. Two-stage targeted minimum-loss based estimation for non-negative two-part outcomes. *Statistical Methods in Medical Research*.
4. **Liu R**, Williams NT, Rudolph KE, Díaz I. General targeted machine learning for modern causal mediation analysis. arXiv preprint arXiv:2408.14620. *Under revision at Journal of American Statistical Association*
5. Wu W, Li F, **Liu R**, Li Y, McAdams-DeMarco M, Geras KJ, Schaubel DE, Díaz I. Leveraging Neural Networks to Profile Health Care Providers with Application to Medicare Claims. arXiv preprint arXiv:2309.15316. 2023 Sep 26. *Under revision at Annals of Applied Statistics*
6. Susmann H, Williams N., **Liu R**, Young J, Díaz I (2025), Computationally and statistically efficient estimation of time-smoothed counterfactual curves. arXiv preprint arXiv:2509.26554. 2025 Sep 30. *Lifetime Data Analysis*
7. Charytan M D, **Liu R**, Wu W, Organic Chemical Values for Patients on Dialysis, *Submitted to Kidney Medicine*.
8. **Liu R**, Abraham R, Conderino SE, Kanchi R, Blecker SB, Dodson JA, Thorpe LE, Charytan DM, McAdams-DeMarco MA, Wu W. COVID-19 Pandemic-induced Healthcare Disruption and Chronic Kidney Disease Progression. *Journal of General Internal Medicine*.
9. Khan NS, Dhanda AK, Takashima M, **Liu R**, Yoshiyasu Y, Wu W, Jin W, McCoul ED, Ramanathan Jr M, Ahmed OG. What is the robustness of randomized controlled trials supporting rhinosinusitis guidelines?. *American Journal of Otolaryngology*. 2025 Jan 1;46(1):104575.
10. Muller, M. A., **Liu R**, et al. (2025). "Clonal Hematopoiesis of Indeterminate Potential in Chronic Coronary Disease: A Report From the ISCHEMIA Trials Biorepository." *Circulation: Genomic and Precision Medicine* 18(3): e004921.
11. Hamo CE, **Liu R**, Wu W, Anthopolos R, Bangalore S, Held C, Kullo I, Mavromatis K, McManus B, Newby LK, Reynolds HR. Cardiometabolic Co-morbidity Burden and Circulating Biomarkers in Patients With Chronic Coronary Disease in the ISCHEMIA Trials. *The American Journal of Cardiology*. 2024 Aug 15;225:118-24.

PUBLICATIONS IN PREPARATION

1. **Liu R**, Díaz I, General Path-Analysis with Multiple Mediators.
2. **Liu R**, Kim R, Stensrud MJ, Santacatterina M, Causal Inference in Crossover Trial.
3. **Liu R**, McClean A, Díaz I, Time-efficient Estimation of Longitudinal Effects.

4. Charytan M D, **Liu R**, Wu W, Association between Dialysis and Stunning for CKD Patients.

TALKS AND CONFERENCES

1. **Presentation** of *General targeted machine learning for modern causal mediation analysis* in SLDS 2026, New York City, New York, USA, Nov 2026.
2. **Organizer** of Session *An Overview of Automatic non-parametric (Riesz) learning for causal effects estimation* in ACIC 2026, Salt Lake City, Utah, USA, May 2026.
3. **Presentation** of *General targeted machine learning for modern causal mediation analysis* in ACIC 2026, Salt Lake City, Utah, USA, May 2026.
4. **Poster** of *General targeted machine learning for modern causal mediation analysis* in ACIC 2025, Detroit, Michiagn, USA, May 2025.
5. **Poster** of *Time-efficient Estimation of Longitudinal Effects* in Annual Poster Competition in the Department of Popu Health, NYU Langone Health, Apr 2026.
6. **Poster** of *General targeted machine learning for modern causal mediation analysis* in Annual Poster Competition in the Department of Popu Health, NYU Langone Health, Apr 2025.
7. **Chair** of Session *Statistical Advances for Complex Event History Data* in ICSA 2024, Nashville, Tennessee, USA, May 2024.

OTHER RESEARCH

Optimize Dictionary Learning Problems by Quasi-Newton Methods in Manifold Optimization Feb 2020 - May 2020

Undergraduate Thesis

- Made numerical stability analysis with Quasi-Newton Methods and tested the speed compared with other Manifold-based methods such as Conjugate Gradient Methods on Manifold.
- Successfully transferred L-BFGS from Euclidean Space to the Manifold Space and ran experiments on different image datasets such as the textile images, achieved approximately twice the speed of the original method.

EDITORS&REVIEWERS

Editor: *Scientific Reports*, *Journal of Open Source Software*.

Reviewer: *Journal of American Statistical Association*, *Journal of Open Source Software*, *Journal of Big Data*, *Journal of Royal Statistical Society: Series C*, *Biostatistics*, *Epidemiology*, *Journal of Causal Inference*, *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*.

PROFESSIONAL EXPERIENCE

Eli Lilly and Company May 2026 - Aug 2026
Ph.D Statistics Intern

New York University July 2023 - June 2024
Asst Research Scientist in Division of Biostatistics, Department of Population Health, NYU Langone Health

AWARDS AND HONORS

- **Best Paper Award** in Annual Poster Competition in the Department of Popu Health, NYU Langone Health, Apr 2024.
- **Student Travel Award** for 2025 Deming Annual Conference, Dec 2025.
- **Student Award** for NESS 2026, Jun 2026.
- **Best Student Honorable Mention** (Top 3), Nov 2022
- **Honored Prize (Admitted in Final, Top 12)** in S.-T. Yau College Student Mathematics Contest, May 2019.
- **National Scholarship**(Top 1% Students in China), September 2017.

ACTIVITIES

Biomedical Data Science (BDS) Club Sep 2024 - Present

Co-Founder

- Provide bi-weekly lectures for BDS topics, including data wrangling (`Pandas` in Python, `tidyverse` in R, etc.), data visualization (`matplotlib, seaborn` in Python, `ggplot2` in R, etc.), modeling (linear regression, logistic regression, longitudinal data analysis, Cox Proportional Hazard model), machine learning&deep learning in Python.
- Invite faculty members for guest lectures, including microbiome data analysis, clinical trial designs, etc...